AI ASSISTED CODING

ASSIGNMENT - 19

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Lab 19

Task-1:

```
class Student:
    def __init__(self, name, age, marks):
        self.name = name
        self.age = age
        self.marks = marks

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)
        print("Marks:", self.marks)
# Test
s1 = Student("Sai", 20, 85.5)
s1.display()
#-------------------------
#include <iostream>
using namespace std;
class Student {
private:
    string name;
    int age;
    float marks;
public:
    // Constructor
    Student(string n, int a, float m) {
        name = n;
        age = a;
        marks = m;
    }
    void display() {
```

```
        cout << "Name: " << name << endl;

        cout << "Age: " << age << endl;

        cout << "Marks: " << marks << endl;

    }

};

int main() {

    Student s1("Sai", 20, 85.5);

    s1.display();

    return 0;

}
```

Output:

```
Name: Sai
Age: 20
Marks: 85.5
```

Task-2:

```
public static boolean isPrime(int n) {

    if (n <= 1) return false;

    for (int i = 2; i <= Math.sqrt(n); i++) {

        if (n % i == 0) return false;

    }

    return true;

}

#----------------------------------------

import math


def is_prime(n):

    if n <= 1:

        return False

    for i in range(2, int(math.sqrt(n)) + 1):

        if n % i == 0:

            return False

    return True

# Test cases
```

```
print(is_prime(2))   # True

print(is_prime(10))  # False

print(is_prime(17))  # True
```

Output:

```
True
False
True
```

Task-3:

```
# FOR i = 0 to n-1

#   FOR j = 0 to n-i-1

#     IF arr[j] > arr[j+1]

#       SWAP arr[j], arr[j+1]

#-------------------------------------

def bubble_sort(arr):

    n = len(arr)

    for i in range(n):

        for j in range(0, n - i - 1):

            if arr[j] > arr[j + 1]:

                arr[j], arr[j + 1] = arr[j + 1], arr[j]

    return arr


# Test

data = [64, 34, 25, 12, 22, 11, 90]

print("Sorted:", bubble_sort(data))
```

Output:

```
Sorted: [11, 12, 22, 25, 34, 64, 90]
```

Task-4:

```
# SELECT name, salary
```

```
# FROM employees

# WHERE salary > 50000;

#-----------------------------

import pandas as pd


# Sample DataFrame

data = {

    "name": ["Sai", "John", "Alice", "Bob"],

    "salary": [60000, 45000, 70000, 30000]

}

df = pd.DataFrame(data)

# Query

result = df[df["salary"] > 50000][["name", "salary"]]

print(result)
```

Output:

```
>> 0    Sai   60000
>> 2  Alice   70000
```

Task-5:

```
def linear_search(arr, key):

    for i in range(len(arr)):

        if arr[i] == key:

            return i

    return -1


arr = [10, 20, 30, 40]

print(linear_search(arr, 30))

#----------------------------------------

#include <stdio.h>

int linearSearch(int arr[], int n, int key) {

    for(int i = 0; i < n; i++) {

        if(arr[i] == key)

            return i;
```

```
    }
    return -1;
int main() {
    int arr[] = {10, 20, 30, 40};
    int n = 4;
    int key = 30;
    int result = linearSearch(arr, n, key);
    if(result != -1)
        printf("Element found at index %d\n", result);
    else
        printf("Element not found\n");
    return 0;
}
```

Output:

```
2
```